The AI Periodic Table

Data Sources → Data Preparation → Representations → Model Learning → Model Families → Interaction Layer → System Integration → Agentic Design → Developer Frameworks → Automation Platforms → Applied AI Systems

KNOWLEDGE FOUNDATION · COLUMNS 1–3 INTELLIGENCE ENGINE · COLUMNS 4–5 INTERACTION & INTEGRATION · COLUMNS 6–7 AGENTIC BUILD LAYER · COLUMNS 8–10 APPLIED AI SYSTEMS · COLUMN 11

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Dc Documents PDFs, webpages, policies, reports. 3.1	**Cn** Cleaning Removing noise and junk. 3.2	**Vd** Vector Database Stores meaning for retrieval. 3.3	**Rl** RLHF Human feedback improves behavior. 3.4	**Gm** Gemini Google multimodal model family. 3.5	**Pe** Prompt Engineering Designing better instructions. 3.6	**Fn** Function Calling Models call tools or code. 3.7	**Pl** Planning Breaking goals into steps. 3.8	**Lg** LangGraph Stateful agent workflows. 3.9	**Mk** Make Visual automation workflows. 3.10	**Dt** Digital Twins Virtual replicas of real systems. 3.11
Ia Images / Audio Multimodal raw material. 4.1	**Ps** Parsing Extracting structure from messy files. 4.2	**Ss** Similarity Search Find close meanings. 4.3	**Ev** Evaluation Measuring output quality. 4.4	**Lm** Llama Open-weight model family. 4.5	**Ce** Context Engineering Selecting the right information. 4.6	**Mc** MCP Model Context Protocol. 4.7	**Tu** Tool Use Agents use external capabilities. 4.8	**Ca** CrewAI Role-based agent teams. 4.9	**Vf** Voiceflow Conversational AI design. 4.10	**De** Digital Employees AI workers for business workflows. 4.11
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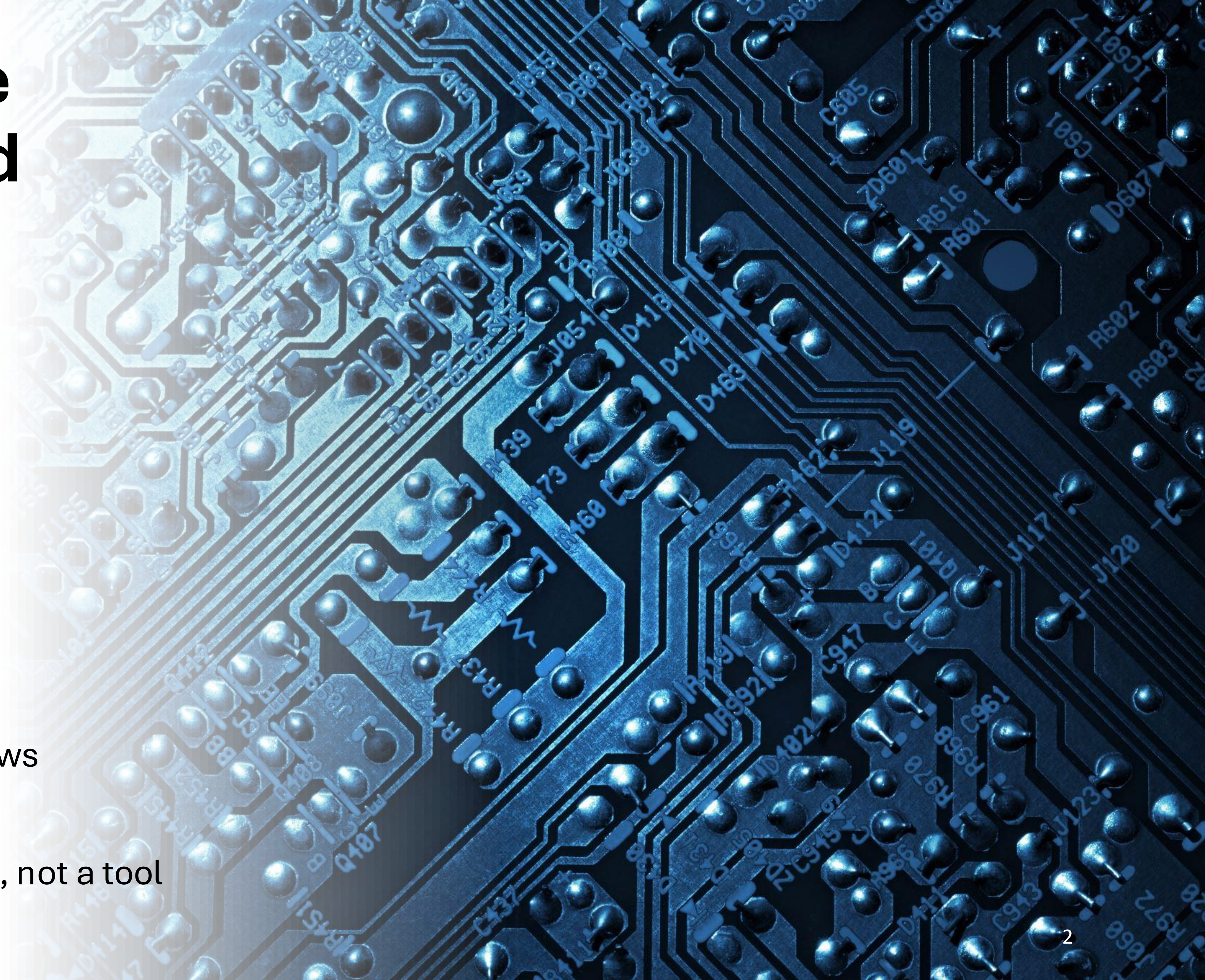
The Last Three Years Changed Everything

2023
- ChatGPT moment
- LLM revolution

2024
- Copilots everywhere
- Multimodal AI

2025
- Agents emerge
- AI integrated into workflows

2026+
- AI becomes a participant, not a tool



THE ERA WE'RE LEAVING BEHIND

2018–2023: The race was about scale — bigger models, more data, higher cost

Training Cost

FROM

~$500

TO

$50–100M+

Parameters

FROM

100M (GPT-1)

TO

~1 Trillion (GPT-4)

Training Words

FROM

~100B

TO

~10 Trillion

Scaling laws worked — until they hit diminishing returns. The next leap isn't bigger models. It's smarter systems.

THE PARADIGM SHIFT

YESTERDAY		TODAY & TOMORROW
Hyperparameter tuning		**Context Engineering**
Model size leaderboards		**Agentic workflows**
Bigger training runs	→	**RAG + memory systems**
One model, one task		**Multi-agent orchestration**
API calls = product		**Autonomous AI systems**

Multi-modal AI

Multimodal Becomes the Default

AI now understands:

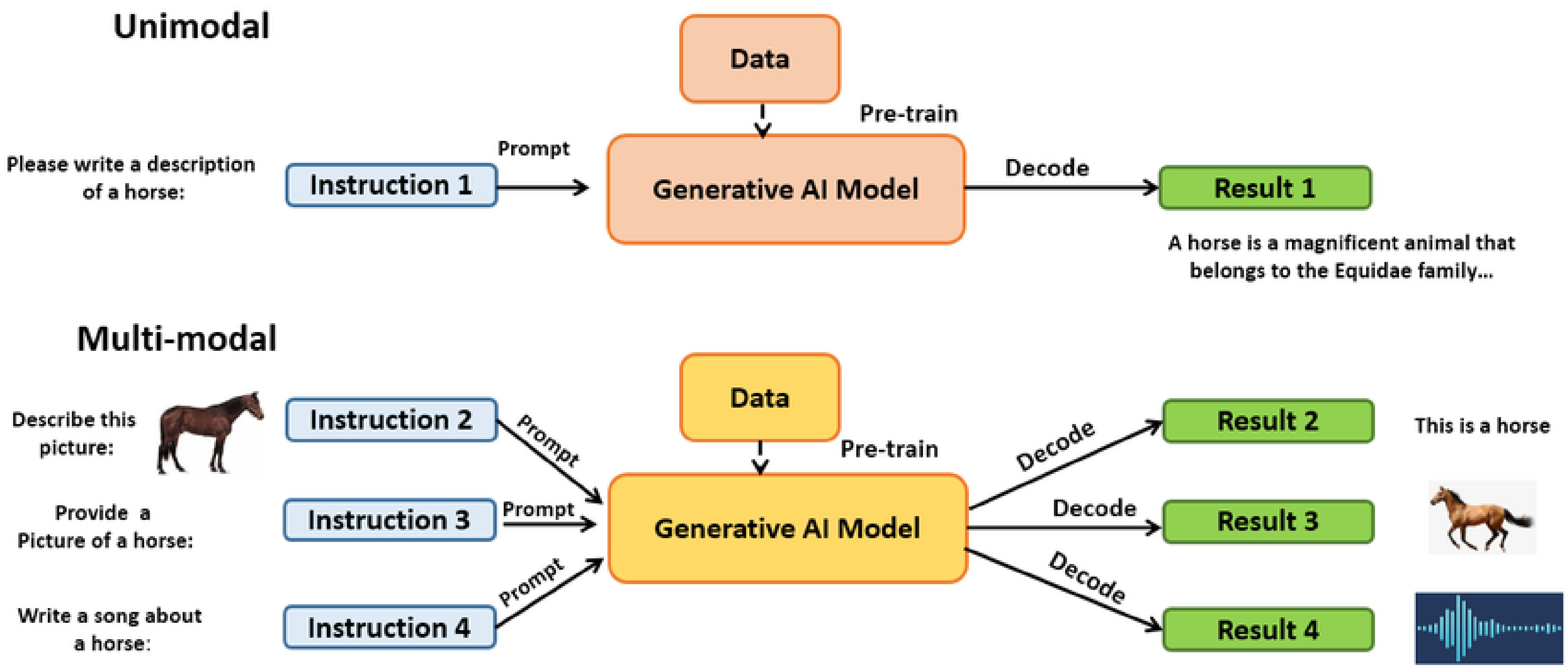
- Text
- Images
- Audio
- Video
- Screens
- Documents

Future:

- Real-time understanding
- Continuous perception
- AI that sees what we see

Examples:

- Medical diagnostics
- Robotics
- Industrial inspection
- Autonomous vehicles
- Digital twins
- Embodied cognition



Unimodal vs Multimodal AI. Source: ResearchGate

Context Becomes the New Intelligence

Old paradigm (a chatbot):

Prompt → Inference

New paradigm (your assistant):

Memory → Context → Reasoning → Action

Topics:

- Long context windows
- Persistent memory
- Enterprise knowledge
- Context engineering

THE AI VALUE PIPELINE

How raw data becomes real-world AI impact

Data Sources	Data Prep	Represen-tations	Model Learning	Model Families	Interaction Layer	System Integ.	Agentic Design	Dev Framework s	Automation Platforms	Applied AI
Knowledge Foundation			Intelligence Engine		Interaction & Integration		Agentic Build Layer			Applied AI Systems

The course mirrors this pipeline: each chapter builds on the last, from raw data to autonomous AI action.

Interactive AI Periodic Table, click it!

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Sn **Sensor Data** Physical-world signals. 6.1		**Rg** **RAG** Retrieval-Augmented Generation. 6.3		**Gr** **Granite** IBM enterprise model family. 6.5	**Ts** **Temperature** Creativity vs predictability. 6.6	**Wh** **Webhooks** Event-triggered communication. 6.7	**Et** **Agent Ethics** Autonomy requires accountability. 6.8		**Cr** **Cursor** AI coding environment. 6.10	**Ao** **AI Operations** Applied AI for business processes. 6.11

KNOWLEDGE FOUNDATION

Td

Training Data

The raw material — text, images, audio, enterprise records.

Tk

Tokenization

Breaking language into model-readable units.

Em

Embeddings

Numbers that carry meaning — the DNA of AI comprehension.

Vs

Vector Space

A geometry of concepts where similar ideas sit close together.

Vd

Vector Database

Where embedding-based memory lives and retrieval begins.

Rg

RAG

Retrieval-Augmented Generation: give the model fresh, relevant context.

RAG: Retrieval-Augmented Generation

The #1 enterprise AI pattern — giving models the right knowledge at the right moment

1 User Query → 2 Embed Query → 3 Vector Search → 4 Retrieve Context → 5 Augment Prompt → 6 LLM Generates

Why RAG Matters for Business

✓ Your data stays private — the model never needs to be retrained on your proprietary information

✓ Answers are grounded in your actual documents, policies, and knowledge bases — reducing hallucinations

✓ Update the knowledge base without retraining — real-time accuracy at enterprise scale

MODEL FAMILIES

You don't need to build a model. You need to know which one to use.

Closed API

Gp

GPT / OpenAI

Industry standard. GPT-4o, o1, o3.

Closed API

Cl

Claude

Anthropic. Strong reasoning & safety.

Closed API

Gm

Gemini

Google. Multimodal powerhouse.

Open Weight

Lm

Llama

Meta. Run locally or fine-tune freely.

Open Weight

Ms

Mistral

Fast & efficient. European sovereignty.

Enterprise

Gr

Granite

IBM. Auditable, enterprise-grade.

CONTEXT ENGINEERING

The most important skill in AI product development today

Prompt Engineering (2022–2023)

× Write clever prompts

× Add 'think step by step'

× Chain a few instructions

× Hope the model figures it out

Context Engineering (Now)

✓ Curate exactly the right information

✓ System prompts define behavior & persona

✓ Memory + retrieval feed live context

✓ Dynamic context window management

The model's intelligence is fixed. What you feed it is your competitive advantage.

AGENTIC DESIGN

AI AGENT

🧠 LLM
The brain

💾 Memory
Short + long term

🔧 Tools
APIs, code, browsers

🎯 Goals
Task + autonomy

Agent Ethics: Autonomy demands accountability. Who is responsible when an agent acts?

MULTI-AGENT SYSTEMS

Specialized AI agents working together — like a well-run team

🎛️ Orchestrator Agent

🔍 Research Agent

📝 Writer Agent

✅ QA Agent

📤 Delivery Agent

Key Frameworks

LangChain

Chain model + tools + retrieval

LangGraph

Stateful, graph-based workflows

CrewAI

Role-based agent teams

AutoGen

Agent-to-agent conversation

SYSTEM INTEGRATION

MCP — Model Context Protocol

A universal standard for connecting AI models to external tools, APIs, and data sources

Function Calling lets models decide when to call a tool. MCP makes any tool reachable.

AUTOMATION PLATFORMS

No-code to pro-code: anyone can orchestrate AI workflows today

Open Source	No-Code	Visual
n8n	**Zapier**	**Make**
Self-hosted workflow automation. Full control, full power.	Connect 6,000+ apps without writing code.	Drag-and-drop visual automation builder.

Conversational	LLM Visual	AI Coding
Voiceflow	**Flowise**	**Cursor**
Design and deploy conversational AI agents.	Open-source visual builder for LLM workflows.	AI-first coding environment. Code with an AI copilot.

APPLIED AI SYSTEMS

Where AI stops being a feature and becomes the business

Digital Employees

AI workers that handle end-to-end business workflows autonomously.

AI Assistants

Enterprise copilots embedded in products, CRMs, and support systems.

Digital Twins

Virtual replicas of real systems for simulation, planning, and testing.

Robotics & Embedded AI

AI controlling physical systems — from manufacturing to autonomous vehicles.

AI Operations

AI managing its own infrastructure — self-healing, self-optimizing systems.

THE ENTREPRENEURIAL LENS

Every layer of the AI pipeline is a business opportunity

Data

Proprietary datasets are defensible moats. Who owns unique data wins.

RAG + Context

Context orchestration services are the new middleware layer.

Fine-Tuning

Vertical-specific models outperform general ones in niche markets.

Agents

Digital workers replace repetitive knowledge work — massive TAM.

Automation

No-code AI tools democratize access — sell to the long tail.

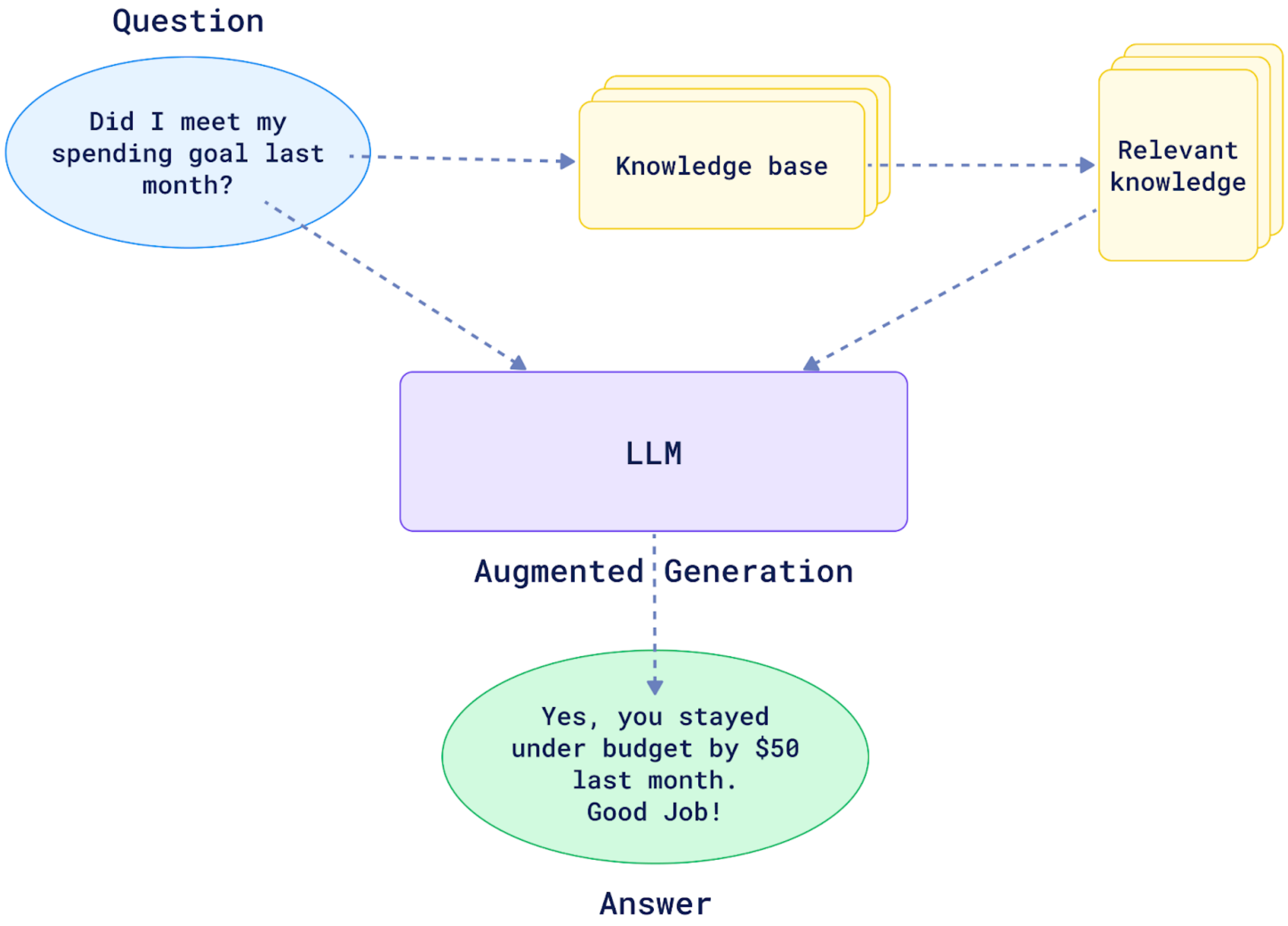
Applied AI

AI-native companies will displace incumbents in every industry.

Td Training Data → Tk Tokenization → Em Embeddings → Gp Model Family → Ce Context Eng. → Mc MCP → Ag Agent → Lc LangChain → N8 n8n → De Digital Employee

Information → Meaning → Intelligence → Action → Impact

The future belongs to those who understand not just what AI can do — but how it works, where it fits, and how to deploy it with purpose.



RAG Evolves into Agentic Knowledge Systems

Old:
- Document search

New:
- Knowledge ecosystems

Concepts:
- RAG
- Agentic RAG
- Graph RAG
- Enterprise memory
- Knowledge networks

AI will increasingly reason across knowledge rather than simply retrieve it.

The Rise of AI Agents

From:

- “Tell me about Araban coffee."

To:

- “Pour me one."

Agent capabilities:

- Planning
- Tool use
- Decision making
- Memory
- Reflection
- Examples:
- Research agents
- Customer service agents
- Software agents
- Business agents

The Digital Workforce

Emerging reality:
•Every employee gains AI teammates.

Examples:
•Digital analyst
•Digital marketer
•Digital researcher
•Digital project manager
•Question for organizations:

How many AI workers will exist per human worker?

Robotics and Embodied AI

Why multimodal matters.
Future systems:

- Humanoid robots
- Autonomous factories
- Smart warehouses
- Healthcare assistants

Key insight:
The physical world is becoming programmable.

Who does what!

Category	Model / Platform	Organization(s)	Description
Foundational transformer lineage	**Transformer architecture**	Google Brain / University of Toronto	The self-attention architecture behind modern LLMs and many vision, audio, and video models. Useful as the conceptual root: tokens become contextual representations through attention.
Foundational transformer lineage	**BERT**	Google	Encoder-only transformer family used heavily for language understanding, classification, search, and embeddings. Still useful historically because it shows the difference between understanding-oriented and generation-oriented models.
Foundational transformer lineage	**T5**	Google	Text-to-text transformer that framed many NLP jobs—translation, summarization, question answering, classification—as one unified text-in/text-out task.
Frontier LLMs / reasoning	**GPT-5.6 family: Sol, Terra, Luna**	OpenAI	Current OpenAI frontier family in the API docs. Sol is positioned for complex reasoning/coding, Terra for balancing intelligence and cost, and Luna for high-volume cost-sensitive workloads. Supports text/image input, multilingual work, vision, tools, and computer-use style workflows.
Frontier LLMs / reasoning	**Claude Fable 5, Claude Opus 4.8, Claude Sonnet 5, Claude Haiku 4.5**	Anthropic	Current Claude lineup in Anthropic docs. Fable 5 is positioned for next-generation long-running agents; Opus 4.8 for complex agentic coding and enterprise work; Sonnet 5 for speed/intelligence balance; Haiku 4.5 for fast near-frontier work.
Frontier LLMs / reasoning	**Gemini 2.5 Pro, Gemini 2.5 Flash, Gemini Flash-Lite**	Google DeepMind / Google	Multimodal Gemini family for reasoning, coding, low-latency tasks, and budget-friendly high-volume work. Gemini 2.5 Pro is positioned for complex tasks; Flash emphasizes price-performance and speed.
Frontier LLMs / reasoning	**Grok 4.5**	xAI / SpaceXAI	General and coding/agentic model family associated with xAI. Useful in a survey table as an example of a high-profile competitor emphasizing real-time information, technical work, and agentic workflows.

Category	Model / Platform	Organization	Description
Open / source-available LLMs	**Llama 4 Scout and Llama 4 Maverick**	Meta AI	Source-available Llama family models used for open-weight experimentation, local or enterprise deployment, and fine-tuning workflows.
Open / source-available LLMs	**Mistral / Le Chat / Magistral family**	Mistral AI	European AI model and assistant ecosystem, including fast chat use, coding, enterprise deployments, and reasoning-oriented models. Often discussed as an alternative to the US frontier labs.
Open / source-available LLMs	**DeepSeek-R1 / DeepSeek-V3**	DeepSeek-AI	Reasoning-focused and efficient LLM family. R1 became influential because it demonstrated strong chain-of-thought-style reasoning behavior and open/distilled variants for research and local use.
Open / source-available LLMs	**Qwen3 / Qwen3 Coder / Qwen Omni**	Alibaba Cloud	Large multilingual model family with dense and mixture-of-experts variants. Useful for teaching global model competition and open-weight alternatives.
Multimodal and image generation	**CLIP**	OpenAI	Classic vision-language model that connected image and text representations through contrastive learning. Historically important because it helped make text-guided image generation more practical.
Multimodal and image generation	**GPT Image 2**	OpenAI	OpenAI image-generation and editing model listed in the specialized models catalog. Used for text-to-image, image editing, design iteration, and visual prototyping.
Multimodal and image generation	**Nano Banana / Nano Banana 2 / Nano Banana Pro**	Google	Google generative media models for image generation and editing. The newer line is positioned for fast creative workflows, production-scale visual creation, and professional design tasks.
Multimodal and image generation	**Midjourney**	Midjourney	Popular image-generation platform known for visually rich, stylized, art-direction-heavy outputs. Useful for marketing concepts, mood boards, storyboards, and creative ideation.
Multimodal and image generation	**Stable Diffusion / Stable Diffusion 3 family**	Stability AI / open ecosystem	Diffusion-based image-generation family with a large community of open tools, fine-tunes, and local workflows. Good example of the open generative-media ecosystem.
Multimodal and image generation	**FLUX**	Black Forest Labs	High-quality image-generation model family used for photorealistic and design-oriented outputs. Often appears in AI art, product mockups, and visual content workflows.
Multimodal and image generation	**Krea**	Krea AI	Creative AI platform for image generation, enhancement, real-time visual exploration, and design/marketing workflows. Good for students who want a visual tool with low technical friction.

Category	Model / Platform	Organization(s)	Description
Agentic systems and coding agents	**OpenAI Agents SDK / ChatGPT Work / Codex**	OpenAI	Agent and coding stack for tool use, orchestration, computer use, files, web search, and professional work. Represents the shift from chatbot answers to AI systems that can take actions.
Agentic systems and coding agents	**Claude Code / Managed Agents / Computer Use**	Anthropic	Anthropic agentic stack for coding, long-running tasks, tool use, enterprise workflows, and computer-use scenarios. Strong example of agentic AI applied to software and business operations.
Agentic systems and coding agents	**Gemini Live / Gemini API / GenAI tools**	Google	Google model ecosystem for multimodal assistants, audio/video agents, Gemini API development, and integration with Google services. Good example of AI inside a productivity ecosystem.
Agentic systems and coding agents	**LangGraph**	LangChain	Framework for building long-running, stateful, graph-based agent workflows. Useful for teaching that agent systems need memory, branching, tools, checkpoints, and evaluation, and not just a single prompt.
Agentic systems and coding agents	**CrewAI**	CrewAI	Multi-agent orchestration framework built around agents, roles, tasks, and crews. Useful for teaching delegation patterns such as researcher, planner, writer, reviewer, and executor agents.
Agentic systems and coding agents	**AutoGen**	Microsoft Research	Agent framework for multi-agent conversations, tool use, coding assistance, and collaborative workflows. Good historical and practical example of structured agent collaboration.
Agentic systems and coding agents	**LlamaIndex Workflows**	LlamaIndex	Framework for building knowledge-heavy LLM applications with retrieval, agents, structured data, workflow steps, and document-grounded responses.
Automated workflow agents	**n8n**	n8n	Low-code, node-based automation platform popular for connecting LLMs with apps, APIs, Gmail, Slack, databases, webhooks, and human approval steps.
Automated workflow agents	**Zapier AI / Zaps / Copilot**	Zapier	No-code automation platform for connecting apps through triggers and actions. AI features help users build, explain, and troubleshoot workflows through natural-language instructions.
Automated workflow agents	**Make / Make AI Agents / Maia**	Make	Visual automation platform built around scenarios. Useful for teaching workflows as diagrams: triggers, routers, filters, actions, retries, and app integrations.
Automated workflow agents	**Relevance AI**	Relevance AI	AI workforce and agent platform for building task-specific agents, workflows, sales assistants, research assistants, and internal automations.

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Vibe-coding / app builders	**Lovable**	Lovable	Natural-language app builder for creating websites and applications from prompts. Strong teaching tool for MVPs, landing pages, dashboards, and rapid product prototyping.
Vibe-coding / app builders	**Bolt.new**	StackBlitz	Browser-based AI app builder that can generate, edit, preview, and deploy web apps from prompts. Useful for showing students the connection between product requirements and working software.
Vibe-coding / app builders	**Replit Agent**	Replit	AI coding assistant inside a cloud IDE that can create, modify, debug, and deploy apps. More technical than Lovable/Bolt, but powerful for moving from prototype to working code.
Vibe-coding / app builders	**Cursor / Composer**	Anysphere	AI-first coding environment for professional developers. Useful in a table as the bridge between beginner vibe-coding and serious software engineering workflows.
Video generation	**Veo 3.1 / Veo 3.1 Lite**	Google DeepMind / Google	Cinematic video-generation and editing model family with creative controls and synchronized audio according to Google's generative media model catalog.
Video generation	**Runway Gen-4**	Runway	Video-generation platform/model family used for cinematic clips, image-to-video, camera movement, visual consistency, and creative production workflows.
Video generation	**Kling AI / Kling 3.x**	Kuaishou	Text-to-video and image-to-video platform from Kuaishou. Useful for showing global competition in AI video and marketing/social-media video creation.
Video generation	**Dream Machine**	Luma Labs	Text-to-video and image-to-video platform known for realistic motion and short cinematic clips. Good example of low-friction video prototyping.
Video generation	**Pika**	Pika Labs	Consumer-friendly AI video creation platform for turning text or images into short videos, visual effects, and social-media-ready clips.
Video generation	**Synthesia / HeyGen**	Synthesia / HeyGen	AI avatar video platforms for training videos, explainers, sales enablement, onboarding, and multilingual corporate video. Useful as a practical business-video category.

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Music, voice, and audio	**Suno**	Suno, Inc.	AI music-generation platform that creates songs with vocals and instrumentation from prompts. Strong example for marketing jingles, class demos, and creative production discussions.
Music, voice, and audio	**Udio**	Udio	AI music-generation platform that creates songs from text prompts and can support remix/extension workflows. Useful for comparing prompt-driven music creation tools.
Music, voice, and audio	**ElevenLabs**	ElevenLabs	Voice AI platform for text-to-speech, voice cloning, dubbing, sound effects, voice agents, and increasingly music/audio workflows. Strong for narration, podcasts, and video voiceovers.
Music, voice, and audio	**Gemini TTS / Flash Live**	Google	Speech and real-time audio models in the Gemini API for text-to-speech, bidirectional voice agents, audio streaming, and voice-first AI applications.
Music, voice, and audio	**Adobe Firefly Audio / Podcast / Descript-style tools**	Adobe / Descript ecosystem	Representative category for AI-assisted audio editing, cleanup, transcription, podcast workflow, captions, and short-form content production.
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Music, voice, and audio	**Gemini TTS / Flash Live**	Google	Speech and real-time audio models in the Gemini API for text-to-speech, bidirectional voice agents, audio streaming, and voice-first AI applications.
Music, voice, and audio	**Adobe Firefly Audio / Podcast / Descript-style tools**	Adobe / Descript ecosystem	Representative category for AI-assisted audio editing, cleanup, transcription, podcast workflow, captions, and short-form content production.

Organizations and links to get in and try them out
• **OpenAI API model catalog:** https://developers.openai.com/api/docs/models
• **Anthropic Claude model overview:** https://platform.claude.com/docs/en/about-claude/models/overview
• **Google Gemini API model catalog:** https://ai.google.dev/gemini-api/docs/models
• **Meta Llama models:** https://www.llama.com/models/
• **Mistral Le Chat product page:** https://mistral.ai/products/le-chat
• **Qwen project / models:** https://qwen.ai/
• **DeepSeek-R1 technical report:** https://arxiv.org/abs/2501.12948
• **n8n:** https://n8n.io/
• **Zapier:** https://zapier.com/
• **Make:** https://www.make.com/
• **LangGraph / LangChain:** https://www.langchain.com/langgraph
• **Lovable:** https://lovable.dev/
• **Bolt:** https://bolt.new/
• **Runway:** https://runwayml.com/
• **Kling AI:** https://klingai.com/
• **Luma Dream Machine:** https://lumalabs.ai/dream-machine
• **Suno:** https://suno.com/
• **Udio:** https://www.udio.com/
• **ElevenLabs:** https://elevenlabs.io/

Final thought

The first wave of AI taught machines to generate.

The second wave teaches machines to act.

The third wave will teach machines to collaborate.

The future belongs to those who understand not just models...

...but systems.